

Jun 14, 2021

Introduction to Computer and Network Security

Part 1

Question 1 [40 points]

Explain how to protect a password file with hashing and salting; discuss (a) what is hashing, (b) what is salting, (c) why hashing is not enough for protecting passwords and how salting mitigates the problem of hashing, (d) describe the structure of a password file, and (e) explain if salts can be stored in clear.

Question 2 [10 points]

What is a digital certificate and what are its main components?

Part 2

Question 3 [40 points]

In the context of access control, explain (a) explain what is a confused deputy with the help of an example, (b) how capabilities allow for avoiding a confused deputy attack, (c) explain the notion of Discretionary Access Control (DAC), (d) explain the notion of Mandatory Access Control (MAC), and (e) discuss the main advantages and disadvantages between DAC and MAC.

Question 4 [10 points]

Describe an SQL injection attack. Which are the main mitigations to an SQL injection attack?

Please, remember...

On each sheet, make sure to write: Name Surname Student ID number Question number	The file containing the scan of your answers should be named as follows: surname-name-studentIDnumber.pdf
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